Regular Expressions

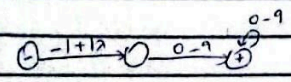
Keywords:
(start|finish|loop|condition|declare|output|input|function|return|break|continue|else)
Identifiers: [A-Z][a-z0-9_]{0,30}
Integer Literals: [+-]?[0-9]+
Floating Point Literals: [+-]?[0-9]+\.[0-9]{1,6}([eE][+-]?[0-9]+)?
String Literals: "([^"\\\n]|\\["\\ntr])*"
Character Literals: '([^'\\\n]|\\['\\ntr])'
Boolean Literals: (true|false)
Operators:

- **Arithmetic:** (**|[+\-*/%])
- **Relational:** (==|!=|<=|>=|<|>)
- **Logical:** (&&|\|\||!)
- **Assignment:** (\+=|-=|*=|/=|=)
- **Inc/Dec:** (\+\+|--)

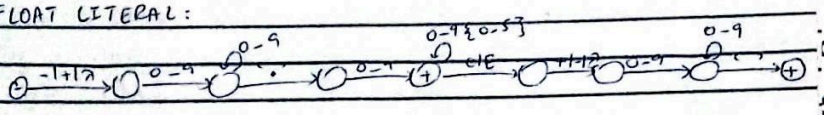
Punctuators: [(){}[\][,;:]]
Single-Line Comments: ##[^\n]*
Multi-line Comments: #*([^*]|*+[^*#])**+#
Whitespace: [\t\r\n]+

NFA

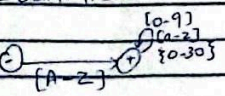
INTEGER LITERAL:



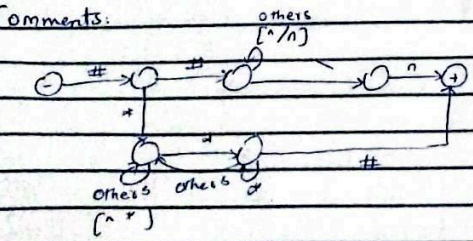
FLOAT LITERAL:



Identifiers:

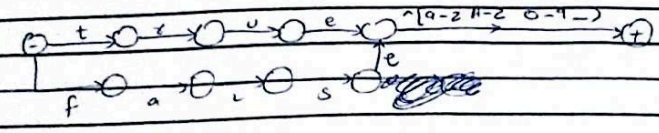


Comments:

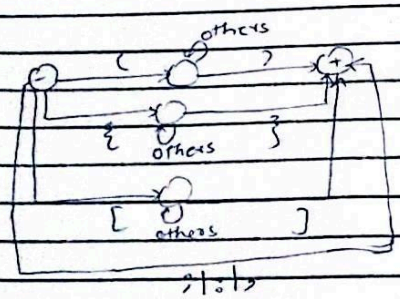


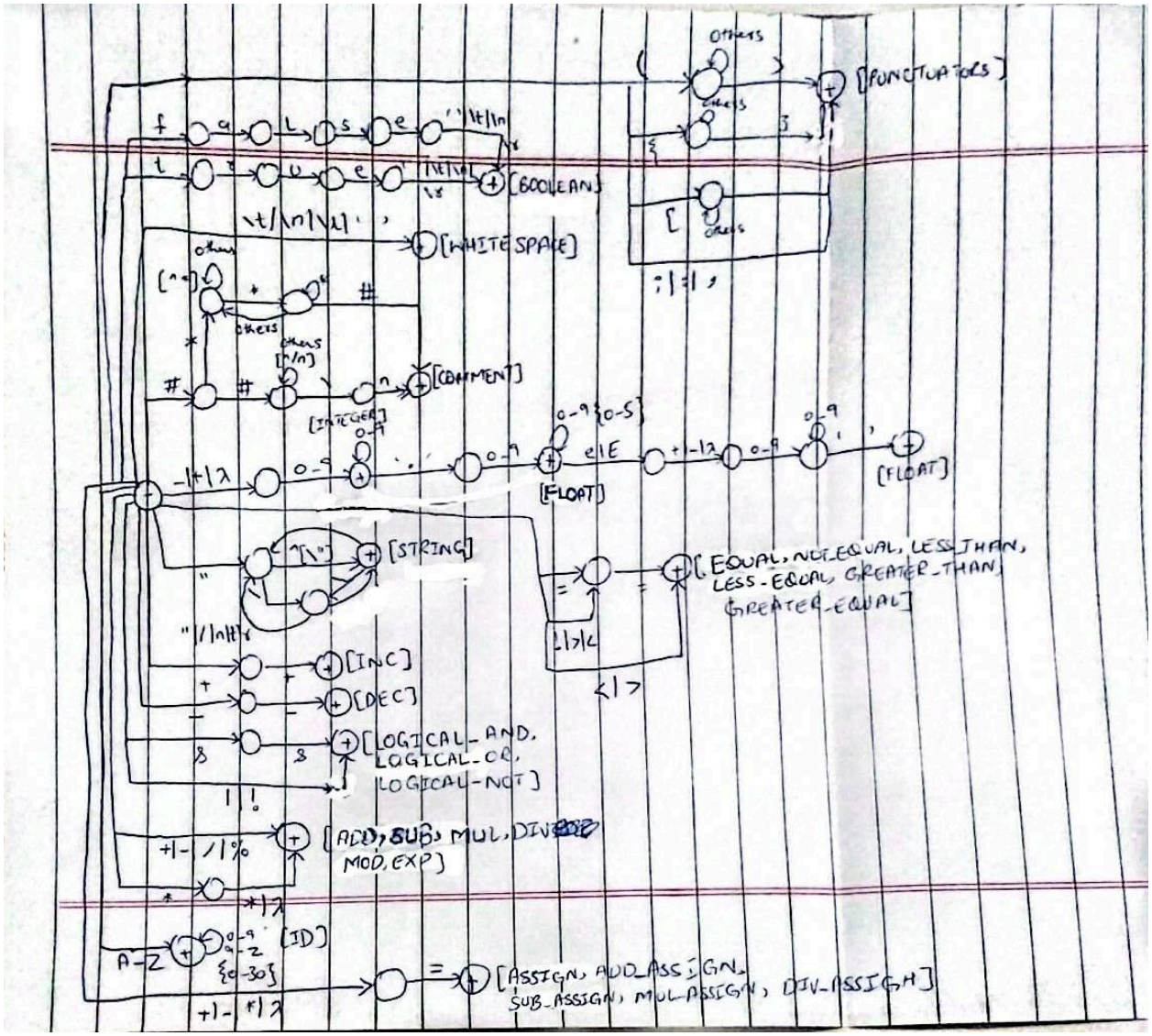
White space:

Boolean:

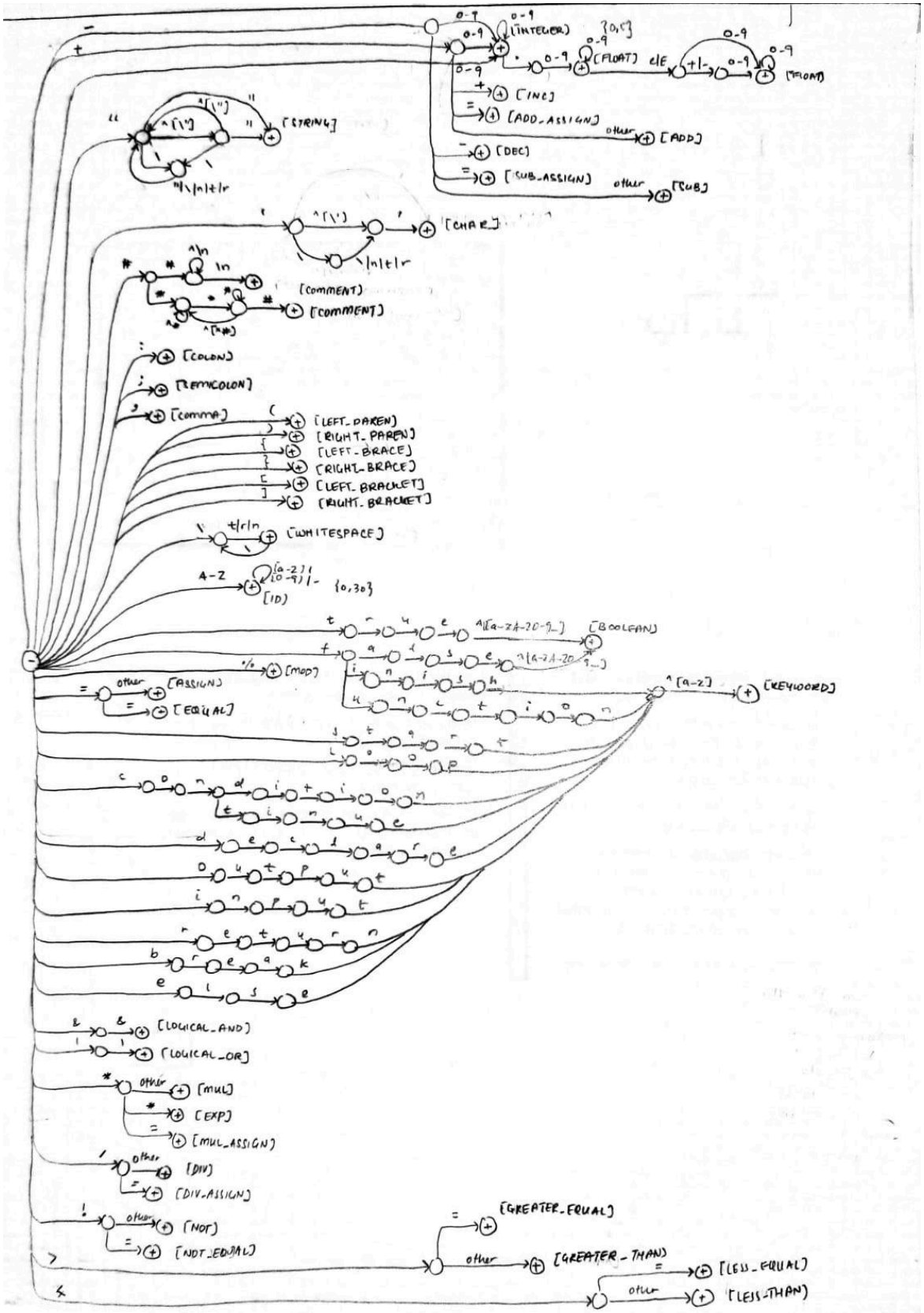


Punctuators:

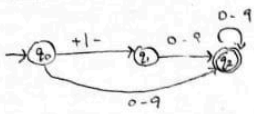



[PUNCTUATORS]
[BOOLEAN]
[WHITE SPACE]
[COMMENT]
[INTEGER]
[FLOAT]
[FLOAT]
[STRING]
[EQUAL, NOT_EQUAL, LESS_THAN, LESS_EQUAL, GREATER_THAN, GREATER_EQUAL]
[INC]
[DEC]
[LOGICAL_AND, LOGICAL_OR, LOGICAL_NOT]
[ADD, SUB, MUL, DIV, MOD, EXP]
[ID]
[ASSIGN, ADD_ASSIGN, SUB_ASSIGN, MUL_ASSIGN, DIV_ASSIGN]

Minimised DFA & Transition Tables

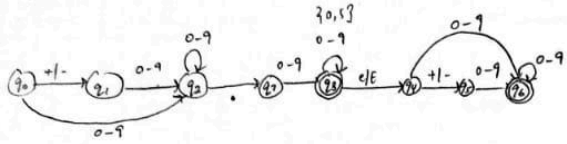


Integer Literal



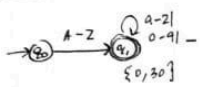
	+	-	0-9	other
q_0	q_1	q_1	q_2	$\emptyset$
q_1	$\emptyset$	$\emptyset$	q_2	$\emptyset$
q_2	$\emptyset$	$\emptyset$	q_2	$\emptyset$

Floating - Point Literal



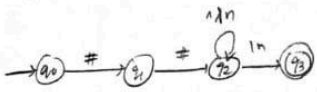
	+	-	0-9	e	E	.	other
q_0	q_1	q_1	q_2	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	$\emptyset$	q_2	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\emptyset$	q_2	$\emptyset$	$\emptyset$	q_7	$\emptyset$
q_3	$\emptyset$	$\emptyset$	q_3	q_4	q_4	$\emptyset$	$\emptyset$
q_4	q_5	q_5	q_6	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_5	$\emptyset$	$\emptyset$	q_6	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_6	$\emptyset$	$\emptyset$	q_6	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_7	$\emptyset$	$\emptyset$	q_3	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Identifier



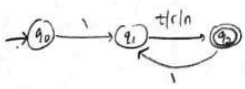
	A-Z	a-z	0-9	_
q_0	q_1	$\emptyset$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	q_1	q_1	q_1

Single Line Comment



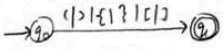
	#	\n	other
q_0	q_1	$\emptyset$	$\emptyset$
q_1	q_2	$\emptyset$	$\emptyset$
q_2	q_2	q_3	q_2
q_3	$\emptyset$	$\emptyset$	$\emptyset$

Whitespace



	\	t	r	n	other
q_0	q_1	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	q_2	q_2	q_2	$\emptyset$
q_2	q_1	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

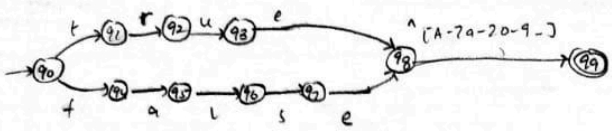
Punctuators



	(	)	{	}	[	]	other
q_0	q_1	q_1	q_1	q_1	q_1	q_1	$\emptyset$
q_1	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

→

Boolean



	t	r	u	e	f	a	l	s	other
q_0	q_1	ϕ	ϕ	ϕ	q_4	ϕ	ϕ	ϕ	ϕ
q_1	ϕ	q_2	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_2	ϕ	ϕ	q_3	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_3	ϕ	ϕ	ϕ	q_8	ϕ	ϕ	ϕ	ϕ	ϕ
q_4	ϕ	ϕ	ϕ	ϕ	ϕ	q_5	ϕ	ϕ	ϕ
q_5	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	q_6	ϕ	ϕ
q_6	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	q_7	ϕ
q_7	ϕ	ϕ	ϕ	q_8	ϕ	ϕ	ϕ	ϕ	ϕ
q_8	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	q_9
q_9	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ